



Course Syllabus

Course Name: Physics Laboratory I

Course Code: PHYS1410J

Course Pre-requisites: Physics I (Vp140) or Honors Physics I (Vp160), concurrently

Textbook: Lab manual, Hugh D. YOUNG, Roger A. FREEDMAN, University Physics (14th edition)

Instructor: WANG Yuxing (wyx75@sjtu.edu.cn)

Office hours: Wed. 9:00-11:00 (No.211 Physics Building)

Course description and detailed teaching schedule:

Physics laboratory PHYS1410J is aimed at providing students with hands-on experience in an introductory physics laboratory.

You will conduct “inquiry” experiments (virtually by observing through videos) where you are not expected to know the outcome in advance. A major goal of this lab course is to facilitate development of your qualitative reasoning skills. During this laboratory centered course you will be exposed to qualitative reasoning skills that scientists use when solving problems such as formulating hypothesis, organizing data, making inferences from data, and designing experiments. Your goal is to learn techniques for analyzing and interpreting data. A key objective is for you to learn how to approach a particular question.

Laboratory Sessions

The entire class will meet once for 2 hours during the 1st week for introduction lecture by Prof. WANG. Thereafter students during the semester, will conduct a total of 5 assigned laboratory experiments, **one experiment per week**. Students are required to read the lab manual and the corresponding sections of the textbook before the session.

- (1) Collisions and Projectile Motion
- (2) Fluid viscosity
- (3) Simple harmonic motion: oscillations in mechanical systems
- (4) Measurement of the speed of sound
- (5) Damped and driven oscillations, Mechanical resonance

Laboratory sessions start from week 1 and will be preceded by an introductory lecture explaining the structure of the course, basic experimental technique, data analysis, report requirements etc. The exact time and information will be announced on Canvas.

Teaching Assistants/Laboratory Sessions Instructors

2023-2024(2)-PHYS1410J 周三、周五 上午8:55-11:40							
	No.	Lab Name	Lab Room	Chinese Name	TA in charge (Phone,Email)		
PHYS1410J Physics Lab1	1	Collisions and Projectile Motion	3#203	碰撞打靶	周乐康	18071757739	zhoulekang@sjtu.edu.cn
	2	Viscosity measurement with Stoke's method	3#411	液体粘滞系数测定	吕彦霖	15821671136	lv10529@sjtu.edu.cn
	3	Simple harmonic motion	3#106	简谐振动	徐逸飞	13917339913	xvfjzn200411@sjtu.edu.cn
	4	Speed of sound using resonance and phase difference	3#415	声速的测定	蒋晨曦	19101678446	jackbaba@sjtu.edu.cn
	5	Damped and driven oscillations	3#110	波尔共振	孙豪	18562886820	sun.hao@sjtu.edu.cn

Please check Canvas regularly for updates.



Grading Policy

Lab Report (basic requirement): $5 \times 18\% = 90\%$

Lab Report (additional discussion, analysis, or application exploration): $5 \times 2\% = 10\%$

Final grading rank:, judged by averaging each experiment score.

Experiments

Each student is required to conduct the experiment through observing the introduction on lab in order to gain any points for that experiment.

Data Sheet

Measurement data must be recorded on a separate data sheet with the report and will not be accepted if the data is recorded with a pencil. The student's name, ID, date, should be clearly indicated.

Lab report

The students are required to write a lab report and submit it within a week after the laboratory session. Reports are due on Friday (**one week after the session**) at 23:30 (UTC/GMT +8.00). The electronic version (in the pdf/word format), needs to be submitted on-line in Canvas by this deadline. You are allowed to submit on-line **only once**, so please make sure that the correct file is uploaded.

Please plan your time well, as late submissions will carry a penalty: 5 score of the marks will be subtracted if the report is submitted after the deadline, 5 score per day of delay will be subtracted till the report is submitted.

Honor Code

Oral discussion of the contents of laboratory exercises with other students is allowed and encouraged. It is not allowed to show any written work to other students. **No electronic data is allowed to be shared** (including raw measurement data and plots). In particular, the students are required to type the lab report, enter the raw data into the data analysis software, and plot all graphs by themselves. Use of fabricated, altered, and other students' data in the report is a violation of the Honor Code.

Report forms provided by the instructor during the laboratory session must not be copied, photographed, or removed from the laboratory room. If any references to academic textbooks or research journals are made, they should be properly identified with bibliographical data. No references to Wikipedia entries are allowed. Any suspected violation of the Honor Code will be reported to the Honor Council.